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Banking AI Assistant For Unsighted People

Using Natural Language Processing

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Abstract: The main objective of banking is to provide the services to the people to manage their money in a safe way. Banking is defined as "Accepting deposits of money from the public for the purpose of Lending or Investment, repayable on demand and withdrawable by cheque, draft, or otherwise. when it comes to the blind people they have to depend on others for daily activities. So, we are making an effort to help the blind in the banking sector which enables them to do physical activities like filling a form, check balance, update details etc on their own. People can use the banking services such as deposit, withdraw, loans, etc on their own. But the scenario when it comes to blind people he/she must have to depend on others to operate the bank account on behalf of the blind person. The bank may accede visually impaired customers request to appoint a person/persons as his/her Power of Attorney or Mandate Holder to operate his/her bank account. If the person is not the trusted one, then the blind people may lose their power on their account where he/she may lose the money or information which is private to his account. We have come up with an innovative solution to solve all the above-discussed problems.

The Rule-based is the simple one when compared to other ones and can respond to customers based on predefined original, human-generated rules. Communication in this form tends to be text-based and often uses on-screen buttons, giving the user a wide choice of responses. These bots can also work by understanding certain keywords and phrases in written communication.

Index Terms – : Banking Bot ; Rule Based ; Artificial Intelligence ; Natural Language Processing .

1. INTRODUCTION

Introducing our innovative banking bot, specifically designed to provide seamless banking experience to the visually impaired. This bot is equipped with state-of-the-art technologies that make it easy for users to access and manage their bank accounts without any assistance. The bot's intuitive interface, voice recognition and text-to-speech capabilities ensure that users can interact with it easily and efficiently. With our bot, blind users can perform a wide range of banking transactions, from checking account balances and updating phone number along with filling forms, all without the need for assistance from a third party.

Our banking bot is a game-changer for the visually impaired, empowering them with the ability to manage their finances with ease and independence.

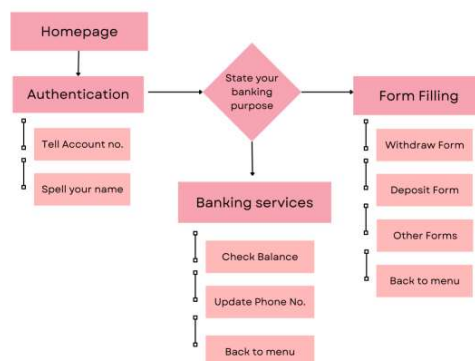


Figure-1 : Flow

The above Flowchart represents the flow of a banking transaction which starts from Homepage representing the Interface with the introduction of the Bot, then there will be the initial authentication of the user to login into the system with their account number. Then the user will be given their choice of choosing the operations of the Form Filling or other banking purposes. Finally the user's purpose will be fulfilled and the application returns to the Homepage.

2. Limitations of Traditional Banking Systems

Traditional methods of banking are generally considered but when it comes to the case of blind people , it's difficult for the transactions to be held .So we are introducing our banking bot to overcome the problem of trust issues.Hence enabling the online transactions for the Unsighted people will resolve the issue.

3.Need of the Banking Bot Application

Our Banking bot can be extremely helpful for visually impaired people, as they provide an accessible and convenient way to manage their finances independently. Here are some ways that a banking bot can help visually impaired people.

4. Technologies to be used in making Banking Bot

To succeed in making the bot different Python Libraries are to be used in different sectors of the project such as:

1. pyttsx3
2. pyaudio
3. speechrecognition
4. OpenCV
5. RASA Algorithm.

These libraries are to be used for speech recognition, audio to text conversion etc.

5. Methodology

5.1 Types of Chatbots

• Rule Based Chatbots

One of the simplest chatbot is rule based chatbot.In these the database of answers and rules are given that help the bot to find the right answer from the given database.very effective and useful answers are created. we can not create our own answers ,but with a large database of answers and well-designed rules.The individual inputs and their responses are stored in the form of arrays in these simplest rule-based chatbot.these bots are very limited and can only answer queries that exactly match the entries provided in the database.

• AI-Based Chatbots

Machine learning is increasing a lot in our recent life ,many new approaches and techniques are developed . For create highly intuitive and precise chatbots artificial intelligence are used that are tailored to specific goals.Using complex machine learning models the AI-based chatbots are designed.

because of these complex machine learning model these chat bot as a capability of learn on their own.

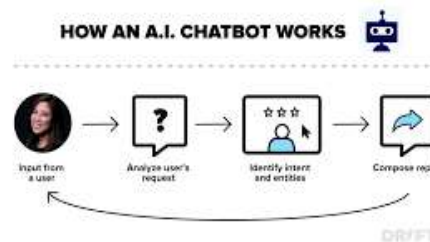


Figure 3:Chatbot Woking

5.2 RASA Framework Working

• Overview

The Rasa framework consists of two components: Rasa NLU and Rasa Core. Rasa NLU is an open source natural language processing engine for intent and entity classification.

It basically processes unstructured user input and extracts the meaning from it. This requires many steps.

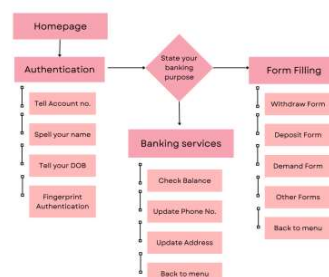


Figure : Working of the System

• NLP(Natural Language Processing)

The combination of the two processes natural language understanding (NLU) and natural language generation (NLG) the Natural language processing (NLP) is made. NLU and NLG are subsets of NLP.

The user input importance are understood by the NLU. Its NLU focuses primarily on machine readability, allowing chatbots to understand the meaning of text. The main work of the NLU is to understand the input text given by the users and it classifies it into appropriate content.

NLU includes:

- Natural Language Inference (NLI) and Interpretation
- Semantic analysis
- Answer the questions
- Sentiment analysis
- Summary of user comments

Human understandable texts are generated by these NLG software. NLG provide the techniques on how to build systems that can leverage the knowledge and capabilities of humans and machines.

NLG involves:

- Content identification
- Discourse planning
- Sentence aggregation
- Lexicalisation
- Referring expression generation

There are two components of an NLP system - natural language understanding (NLU) and natural language generation (NLG). When you enter text into the NLP engine, the user's meaning or context is decoded by the NLU structure and the response is generated by the NLG.

• OpenCV

OpenCV is a Python library that allows you to perform image processing and computer vision tasks. It provides a wide range of features, including object detection, face recognition, and tracking. OpenCV helps us to compare two finger prints whether they are same or not. It uses an algorithm known as SIFT (Scale-Invariant Feature Transform)

6. Capabilities of the Banking Bot

Voice recognition: Banking bots are equipped with voice recognition technology, which allows visually impaired users to interact with the bot using their voice. They can ask questions, give commands, and carry out banking services without needing to see a screen or type.

Text-to-speech capabilities: Banking bots can also read out the information on the screen, such as account balances and other account details, using text-to-speech capabilities.

Intuitive interface: Banking bots are designed with an intuitive interface that is easy to navigate using simple voice commands. This means that visually impaired users can perform a range of banking transactions.

Availability: Banking bots are available, which means that visually impaired users can access their accounts and perform banking transactions at any time.

6.1 The Role Of Text to Speech -: One of the implementations of the Banking Bot includes the conversion of the text to speech, where the Data of the bank is to be spoken out by the bot to help the user listen to the given options. pyttsx3 is a text-to-speech conversion library in Python. It works offline and is compatible with both Python 3. An application invokes the pyttsx3.init() factory function to get a reference to a pyttsx3 library. It is a very easy to use tool which converts the entered text into speech.

6.2 The Role Of Speech Recognition -: With the aid of speech recognition technology, a computer can comprehend spoken words. It is sometimes referred to as speech-to-text or automated speech recognition. Python's speech recognition software can then be used to translate the spoken words into text, ask a question, or respond. Even some equipment can be programmed to react to these spoken words.

6.3 The Role Of OpenCV-: OpenCV is a free and open-source software library for computer vision. As a part of security of the banking transactions, The Face of the User will be detected throughout the transaction processing, and the OpenCV library is used for the face detection.

7. Evaluation Metrics

7.1 Metrics : The metrics to be considered for the application includes the calculation of the data validation, and the accuracy of the recovered data from the server.

$$\text{Accuracy} = \frac{\text{Correct prediction}}{\text{Total cases}} * 100\%$$

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} * 100\%$$

7.2 Metrics Outcome : The accuracy of the system purely relies on the input given to the banking bot, so it may vary from person to person. Along with the accuracy of the bot, there will also be a calculation of the confidence of the system. This refers to the complete application. And unlike the accuracy, there is a calculation of the confidence of the application which will also vary from person to person.

```

[5 rows x 7 columns]
listening...
result2:
[{'alternative': [ {'confidence': 0.79623767, 'transcript': '123 456 0123'},
                  {'transcript': '123 4560 123'},
                  {'transcript': '12345 60 123'},
                  {'transcript': '123 45 60 123'},
                  {'transcript': '12345601 23'}],
  'final': True}]
123 456 0123
1 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
2 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
3 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
4 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
5 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
6 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
0 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
1 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
2 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
3 ['1', '2', '3', ' ', '4', '5', '6', ' ', '0', '1', '2', '3']
1234560123 -class object

```

7. Result and Experimental Analysis

We ran the experiments of our Banking Bot on PCs with Windows 11 Operating System, Intel Core i5-9500E @4.5GHz CPU with 8GB RAM. The utilization of the Banking Bot application has proved to be useful in terms of ease of banking, security throughout the transaction, and other similar banking features. The main functional category of the application is the Form Filling Feature which will be done with the complete security and also the output will be received as a PDF format and the minor features includes the updation of phone number, & also the Balance Checking is also included with the application. The security of the transaction includes the face recognition of the user. The face will be detected throughout the transaction whether it's a major or minor feature. If the face is not recognised there will be an error message displayed as Unknown User. This Security feature uses the pre registered face linked to that account. Hence the combination of most of the banking features including the required security features, results in The Banking Bot Application.

Withdrawl Form
Account No: 1234560123
Account Holders Name: Abhiram
Amount: 2500

Example Output for Withdrawal Form

Deposit Form
Account No: 1234560123
Account Holders Name: Abhiram
Amount: 3000

Example Output for Deposit Form

8. Scaling Features and hyperparameter frames

The Rasa framework is made up of the components Rasa Nlu and Rasa Core. For intent and entity classification these Rasa NLU is used as an open-source natural language processing engine. It basically processes unstructured user input and extracts meaning from it. There are many steps involved in this. Tokenization, vectorization, POS tagging, intent classification, etc. is a number of steps to name a number. Rasa Core is a machine learning framework that acts as a dialog management model. It takes structured input from the Rasa NLU and using a machine learning algorithm it predicts the output to display. Simply put, the Rasa NLU is used to interpret the user's unstructured input and convert it into structured data. Rasa core is used to decide the most likely response or action. Rasa NLU and Rasa Core are independent of each other and can be used separately if needed.

Understanding NLP (Natural Language Processing)

Natural language comprehension (NLU) and natural language generation (NLG) are two mechanisms that are combined in natural language processing (NLP). NLP is a superset of NLU and NLG. NLU understands the importance of user input. Focusing primarily on machine readability, this NLU allows chatbots to understand the meaning of text. NLU basically understands the text input given to it and classifies it into appropriate content.

NLU includes:

- Natural Language Inference (NLI) and Interpretation
- Semantic analysis
- Answer the questions
- Sentiment analysis
- Summary of user comments

NLG is software that generates human-understandable texts. Ideas for creating systems that can utilize the skills and expertise of people and computers are provided by NLG techniques.

NLG involves:

- Content identification
- Discourse planning
- Sentence aggregation
- Lexicalisation
- Referring to generation
- of expression

Natural language understanding (NLU) and natural language output make up an NLP system (NLG). The NLU structure understands the user's purpose or context when text is entered into the NLP engine, and the NLG then generates the appropriate response.

9. Conclusion

To conclude, Banking bot is helpful in guiding blind with correct and most up-to-date sources of information. It is advantageous for Visually impaired people for queries such as balance enquiry, Phone number update, filling various kinds of forms like withdraw from, deposit form. Customers can get the information at their fingertips rather than taking help from some other people. It improves efficiency by taking over tasks for which humans are not essential. This project talks about the capability of chatbots to help the blind people to access basic services of the bank.

This can give a feeling of interaction with humans and can make it a joyful conversation . By implementing the above functionalities, we have added some basic functions of chatbots for the banking services.

We are making our model using NLP, AI, Speech Recognition and ML algorithms that are useful for visually impaired people to access the bank on their own without any external help. chatbot that can interact with visually impaired users in a way that is accessible and user-friendly. Forms are filled by the chatbot by recognising the speech of the user. For this purpose we used open CV , speech recognition . Ai protected security is implemented.

10. Future Scope

In today's time , more and more functions are turning into a digital mode. Which also includes the time to time updation of the software is much required in terms of utility, and security. Which includes the future work as below.

1. Fingerprint Authentication at the initial stage of the transaction.
2. Availability of different languages included for regional convenience.

11. Authors' Note

The authors have disclosed that they have no conflicts of interest that may have influenced their study, and they have assured that their research work is authentic and does not contain any copied content.

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